

Users' contributions to radical innovation: evidence from four cases in the field of medical equipment technology

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This paper focuses on contributions of users in early phases of radical innovation projects. In a multiple case study analysis in the field of medical equipment technology, we identify characteristics of users who contribute substantially to the development of radical innovations by being their inventors and (co)-developers. These innovative users have high motivation to seek new solutions, possess a diverse set of competencies, and are embedded in a supportive environment. We furthermore observe that they play an entrepreneurial role as they establish and organize the required innovation networks. These innovation networks are needed to transform the users' radically new concepts into first physical prototypes and marketable products. The study highlights how manufacturing firms can benefit from innovative and entrepreneurial users in the early phases of radical innovation projects.

1. Introduction and overview

In today's environment of rapid technological change, radical innovations (RI) have become crucial for the long-term competitiveness of firms. Although firms have become increasingly sophisticated in the development of incremental innovations, many companies struggle to generate RI. In order to enhance their innovation capability for RI, they must involve capable actors within and outside the firm (Chesbrough, 2003).

This paper focuses on users of products or technology as innovative entities outside manu-

facturing firms.¹ There is widespread consensus among practitioners and scholars alike that users can provide valuable inputs in the early phases of innovation projects (e.g. Foxall, 1989; Gruner and Homburg, 2000; Cooper et al., 2002; Kristensson et al., 2004). This consensus holds for incremental innovations; however, for RI, two contrasting viewpoints have been advanced. Some scholars doubt whether users can deliver valuable contributions for the early phases of RI projects, but other scholars argue that specific users can contribute substantially. Sceptics contend that users are unable to abstract from their current

use context, and thus are notoriously lacking in foresight (Hayes and Abernathy, 1980; Hamel and Prahalad, 1994; Martin, 1995). Furthermore, they argue that users are cognitively overstrained by the high degree of newness of the technology, particularly in high-technology fields (Moriarty and Kosnik, 1989; Lynn et al., 1996; Veryzer, 1998a; O'Connor and Veryzer, 2001). In contrast, advocates argue that there is a group of users – namely 'lead users' – who experience needs significantly earlier than others and is therefore able to anticipate emerging markets. Thus, advocates propose that lead users can direct manufacturing firms to RI (von Hippel, 1986, 1989; Urban and von Hippel, 1988; ; Herstatt and von Hippel, 1992; Lilien et al., 2002).

Despite a large body of research on RI and user-driven innovation, relatively little is known about whether and how users can contribute substantially to the early phases of RI projects and about the characteristics of capable users. Thus, in our study, we address the following research questions:

- (1) Are users capable of contributing substantially to the early phases of RI projects?
- (2) If users are capable, in what form do they contribute?
- (3) If users are capable, what are the characteristics of these capable users?
- (4) How can manufacturing firms benefit from capable users?

An understanding of these questions is crucial for manufacturing firms that seek to leverage the creative potential of users in order to enhance their firm's RI capability. It is also of importance for policymakers who wish to foster RI.

Our paper combines the research streams of RI with that of user-driven innovations and sheds light on users as a driving force for RI projects. It contributes to innovation research in three ways. First, it identifies characteristics of users who are in a position to be inventors and (co)-developers of RI. Second, it sheds light on the dynamics of the user–manufacturer interface for cases in which users originate RI projects. Third, it provides insights into how manufacturing firms can benefit from innovative users in the early phases of RI projects.

The paper is organized as follows: first, we review the literature related to our topic. Next, the applied research approach is introduced, followed by a presentation of brief case histories of the four selected RI projects and an in-depth analyses of the observed phenomena. Finally,

implications for innovation research and corporate practice are discussed.

2. Literature review

In order to analyze the research questions addressed in our study, the term 'radical innovation' needs to be characterized. In general, RI involves new products or services with a high degree of innovativeness.

Recent state-of-the-art reviews show that *innovativeness is best understood as a multidimensional phenomenon* (Green et al., 1995; Schlaak, 1999; Avlonitis et al., 2001; Danneels and Kleinschmidt, 2001; Hauschildt and Schlaak, 2001; Garcia and Calantone, 2002; Gatignon et al., 2002; Tidd and Bodley, 2002; Salomo, 2003; Dahlin and Behrens, 2005) that relates to *technology*, *market*, and *organizational change*, as well as *environmental alterations*. An innovation is radical in the *market dimension* if it satisfies unmet needs for the first time, resulting in a quantum leap in customer value. The new product may require potential consumers to undergo considerable attitudinal and behavioral change, precipitating customer inertia. A completely new market may be created; however, market risk is also high because large market investments are needed and many as yet unknown competitors may enter the market. The innovation can be called radical in the *technological dimension* if *knowledge* about the product architecture or its components differs significantly from existing knowledge (Afuah, 1998). Existing knowledge often becomes obsolete. The innovation often relies on completely new technological principles, new architectures, or new materials. Innovativeness in the *organizational dimension* relates to the internal change experienced by the innovating organization. Changes may be required in strategy, structure, processes competences, incentive systems, or culture. Organizational innovativeness also refers to the firm's possible inexperience in the field of the specific innovation and to the organizational risks that it faces. In addition to the dimensions of technology, market, and organization, we suggest a fourth dimension relating to such *environmental alterations* as the required establishment of a new infrastructure (e.g. hydrogen filling stations), to regulatory changes, or to changes of societal value systems.

With respect to innovative actors outside of manufacturing firms, users have been shown to be a major source of invention and, ultimately, of innovation (von Hippel, 1988, 2005). A review of

empirical studies reveals that some users contribute substantially to innovation projects by playing the roles of inventors and/or (co)-developers. This phenomenon was proven in numerous and diverse areas such as scientific instruments (Riggs and von Hippel, 1994), CAD software (Urban and von Hippel, 1988), pipe hanger hardware (Herstatt and von Hippel, 1992), surgical equipment (Luethje, 2003), library information systems (Morrison et al., 2000), and sports equipment (Franke and Shah, 2003; Hiennerth, 2004; Franke et al., 2005; Luethje et al., 2005). The pursuit of tailored solutions for their needs motivates users to engage in innovative activities (von Hippel, 2005). User-driven innovations tend to have a low-to-medium degree of radicalness (Luethje, 2003; Luethje et al., 2005; von Hippel, 2005).² Scholars in the field of user-driven innovation, however, have emphasized the potential of a specific group of users – called 'lead users' – to generate RI (von Hippel et al., 2000; Lilien et al., 2002). According to von Hippel (1986; 1989), lead users differ from ordinary users in two respects: lead users face needs months or years before the mass of the marketplace encounters them (trend leadership) and they benefit significantly from obtaining a solution to those needs, and are therefore highly motivated to engage in innovative endeavors. Empirical studies³ on the involvement of lead users in manufacturers' innovation processes reveal that these users contribute substantially to the development of highly innovative and commercially attractive products (Herstatt and von Hippel, 1992; von Hippel et al., 2000; Olson and Bakke, 2001; Herstatt et al., 2002; Lilien et al., 2002). It is important to note, however, that the application of the lead user method does not guarantee the generation of such attractive innovations, because problems that can occur in each phase of the lead user method and process – if no trends can be identified, for example, or if a deficient identification of lead users occurs (Brockhoff, 2003; Luethje and Herstatt, 2004). These problems can even lead to the termination of a lead user project. As the cited lead user studies exclusively report findings of completed lead user projects they might have a positive bias towards the outcomes. Furthermore, the degree of innovativeness of the solutions which were developed in the cited lead user studies was assessed by only one or two indicators. It is therefore difficult to evaluate the extent to which these solutions match our definition of RI.

A comprehensive case-study-based research program on RI, conducted by scholars at the

Rensselaer Polytechnic Institute, draws a more pessimistic picture of substantial user contributions in RI projects. The scholars focused on the management of RI in large, established firms and identified a passive and sometimes even counter-productive role of users (Lynn 1996; O'Connor, 1998; Veryzer, 1998a, b; McDermott, 1999; Leifer et al., 2000; O'Connor and Rice, 2001; McDermott and O'Connor, 2002). In these studies, the RI process followed a common pattern. Large established manufacturing firms were the source of RI and engineers of these firms developed the ideas, technologies, and prototypes. Thus, the locus of invention and development was inside the manufacturing firms. Users were primarily involved at the prototype stage in order to gain market-oriented evaluations and to assess the market potential of the prospective new products. Consequently, users remained as passive residual actors in the RI process. In this connection, it was shown that current users, potential users, and opinion leaders were not able to provide valid evaluations of radically new concepts or prototypes (Lynn et al., 1996; Veryzer, 1998a). However, the passive role of users identified in these studies may be partly because of the chosen focus of the scholars. They focused on large, established manufacturing firms and their challenges in the process of RI projects. Research designs that focus exclusively on the management of RI in large, established manufacturing firms may not fully capture users as contributors. Assuming that users are capable of contributing to RI projects, there are several barriers within established firms that might hinder them from adopting users' contributions. First, ideas and concepts from users may become victims of the 'not invented here' syndrome and therefore be rejected (Vandermerve, 1987). Second, the realization of users' initiatives toward RI tends to destroy the capabilities of established firms because the RI usually requires completely new technologies and resources (Chandy and Tellis, 1998; Shane, 2001). Finally, bureaucratic structures (Chandy and Tellis, 2000) and over-reliance on the voice of mainstream customers (Bower and Christensen, 1995; Christensen, 1997) can hinder established firms from adopting a user's radically new concept.

To recapitulate, scholars who have engaged in lead user research highlight the potential of lead users to make substantial contributions towards RI. However, relatively little is known about the specific competencies, contextual factors, and behaviors of lead users that place them in this favorable position. In contrast to researchers who

focus on lead user research, scholars who focus on the management of RI in large, established manufacturing firms identified a passive and sometimes even counterproductive role of users. Studies on the implementation of the lead user method may have a positive bias towards the role of users, whereas studies that focus on RI within large, established firms may have a negative bias towards users.

In order to better understand the mechanisms that support RI at the user–manufacturer interface, we conducted an in-depth investigation in a high-technology industry, using detailed case studies. The research approach applied is outlined in the next section.

3. Research approach

Instead of focusing *ex ante* on large, established manufacturing firms or on lead user projects within manufacturing firms, we chose a different research approach: we identified RI projects in which users contributed substantially in the early conceptual phases. We then traced back to describe and analyze the roles of these users, their characteristics, and the dynamics that occurred at the user–manufacturer interface.

3.1. Multiple case-study analysis

We conducted a multiple case-study analysis, a methodology suited to the applied focus of our research questions and one that is appropriate, given the dearth of knowledge available in this field of research. Case-study research is especially appropriate for research into new topics and new technologies (O'Connor, 1998; Perry, 1998; Gillham, 2000; Shane, 2000; Stake, 2000; McDermott and O'Connor, 2002), for studies focused on understanding the 'how' and 'why' questions concerning a contemporary set of events, and for studies intended to develop theory further (Eisenhardt, 1989). Multiple cases are generally regarded as being more robust than single case studies because comparisons across cases foster greater validity in the development of insights and fuller consideration of the context dependency of the case project (Yin, 1994; McDermott and O'Connor, 2002).

3.2. Sample and data collection

In order to study our research questions, we sought to include in our sample RI projects in

which users made substantial contributions at the early phases of the project. We needed, therefore, to find a research setting in which RI have been emerging and in which users engage in innovative activities. The area of medical equipment technology was selected, as it fulfills both criteria. In this industry sector, a number of RI emerged in recent years as new communication and information technologies found their way into the operating room. Furthermore, prior empirical studies show that users (medical doctors) play an important role in new product development in this industry sector (Shaw, 1985; Biemans, 1990, 1991; Luethje, 2003).

We focused on actual innovation projects as the unit of analysis and applied a multi-case comparison methodology. For the selection of cases, the following criteria were defined:

- (1) to select RI, the previously mentioned four-dimensional conceptualization of the degree of innovativeness was applied (see Appendix 1), ensuring that we selected RI from both the user' and the manufacturer' perspective; and
- (2) to gain an in-depth understanding of the characteristics of users who are in a position to contribute actively to RI, cases in which preliminary interviews and medical publications indicated that users had played an active role in the development of the RI were selected.

In making the selection, a two-stage filtering process was applied. The first stage comprised nine expert interviews using a semi-structured interview guideline to identify medical disciplines in which RI had occurred frequently within the last 5 years and in which users exhibited intensive innovative activities. Stage one included visits to medical conferences that focused on the application of completely new technologies. In addition, medical journals were scanned for examples of new technologies that are emerging in the operating room. The result of the first stage was a sample of twenty innovations that roughly matched the characteristics of RI. In the second stage, interviews with users, industry experts, and manufacturers were conducted in order to gain preliminary information about the role of users in the innovation process, and to assess systematically the degree of innovativeness of these innovations. Informants assessed the degree of innovativeness of the new products on a seven-point Likert scale. This scale contained a market dimension, a technological dimension, an organizational dimension, and an environmental dimension.

sion (see Appendix 1). Innovations selected for the sample included only those that matched the characteristics of RI by exhibiting a high degree of innovativeness on all four dimensions. This selection procedure ensured that we selected innovations that were also completely new from the perspective of the manufacturing firms. To control for the memory biases of informants, we chose only projects in which innovations were introduced to the market within the last 6 years.⁴ Our multi-stage filtering process resulted in the choice of four RI projects in which preliminary research indicated that, in the early phases of the projects, users had played an active role. The final sample includes a medical robot system, a computer-assisted navigation system for neurosurgery, a computer-assisted navigation system for orthopedics, and a radically new biocompatible implant. The corresponding medical equipment manufacturers are one large established firm, one medium-sized firm, and one spin-off of a large German research establishment. Table 1 provides an overview of the selected RI projects.

Each firm was visited for 2 weeks in order to collect the required data. In-depth interviews, using a semi-structured interview guideline, were conducted with users, on the one hand, and marketing, R&D, project leaders, and executive officers of manufacturing firms on the other. The semi-structured interview guideline for users included questions aimed at learning their roles in the innovation process, their motivation for innovative endeavors, their special knowledge and experience, supportive contextual factors, interaction patterns with partners and manufacturing firms, the degree of innovativeness of their concepts, and the dynamics of the process. The semi-structured interview guideline for manufacturing firms included questions that assessed the innovations in various stages of the innovation process (degree of uncertainty on the market and technol-

ogy dimension, and fit with strategy and core competencies), the role of the firm in the innovation process, the adoption decision and adoption process (why and when the firm had adopted the user innovation), the degree of innovativeness of the new concepts, the market and technological success of the innovations, the users' characteristics, roles, and contributions, and the knowledge transfer and learning effects at the user-manufacturer interface. In total, 36 interviews were conducted. Interviews lasted for 2–3 hours. Each interview was recorded on tape and transcribed.

3.3. Data reduction and coding

A content analysis framework (Neuendorf, 2002; Krippendorff, 2004) was applied in coding the data, whereby category systems were developed for user characteristics, user activities and corresponding roles, interaction patterns between users and manufacturing firms, and learning effects for manufacturing firms. Categories were developed deductively (based on existing theories and concepts) and inductively (derived from the collected data). The deductive section was constructed before the conduct of the interviews. User innovation theory, theories on creativity, and diffusion theory served as bases for the deductive component of user characteristics.

Categories of user characteristics included the following:

- Motivational factors (unsolved problems, intrinsic motivation, openness to new technologies), competencies (special experience and knowledge in the user domain, technological knowledge), and contextual factors (resources for research endeavors, access to interdisciplinary know-how).

Pre-defined categories for user roles were as follows:

Table 1. Selected RI.

Case	Product description	Innovation success	Number of Interviews
URS	Robotic system for neurosurgery	MS: failure TS: high	8
SPOCS	Computer-assisted navigation system for neurosurgery	MS: medium TS: high	9
OrthoPilot	Computer-assisted navigation system for orthopaedics	MS: high TS: high	10
IMPLANT (acronym)	Biocompatible implant	MS : high TS: high	9

MS, market success; TS, technological success; URS, Universal Robot Systems; SPOCS, Surgical Planning and Orientation Computer System.

- user as inventor and
- user as (co)-developer. The development contributions of users were differentiated according to an activity dimension (passive: mere articulation of needs, problems, and evaluations of prototypes, versus active: development of one's own solutions) and a domain dimension (user domain, technological domain).

Regarding interaction between users and manufacturing firms, we analyzed the type of interaction (personal interaction versus non-personal interaction), interaction frequency, and interaction duration. Finally, in recording learning effects for manufacturing firms, our deductive coding scheme included categories such as the type of knowledge gained from users (technological and/or need- and use-related knowledge), acquisition of ideas/concepts, impact on product quality, user-friendliness, development time and cost, and decision quality. The inductive component of the coding system reflects the exploratory nature of the study. Because we were researching a new topic area, we expected to discover patterns and factors not yet discussed in the literature. Each category in the coding scheme was specified with several indicators (see Appendix 2 for selected coding tables). The category system was applied exclusively to the transcribed interview protocols. Expert interviews and archival data were used for verification.

The technological and market success of the RI projects was evaluated by executive officers of the manufacturing firms using a seven-point Likert scale. Market success was evaluated by the following indicators: goal achievement with respect to profits, competitive advantage, customer satisfaction, customer acceptance, and an overall item. Technological success was assessed by asking whether the RI met the technical specifications. Although more commonly used in quantitative studies, Likert scales were incorporated into this qualitative study in order to provide a more detailed assessment of the degree of innovativeness and innovation success.

3.4. *Drawing conclusions and verifying data*

To control for informant bias that might result from the different corporate functions or hierarchical levels of informants, only those statements with a high degree of agreement among informants were included in the analysis. Triangulation of data was applied to validate the

information from the interviews (Maxwell, 1996; Amaratunga and Baldry, 2001). We used several sources of verification. Interviews with three experts from the medical equipment industry were conducted using a semi-structured interview guideline. The guideline topics focused on key issues of our study such as the roles of users and manufacturers in the selected RI projects, the characteristics of users, and the ultimate success of the innovations. Archival data were used in order to obtain quasi-objective data. In this connection, we screened the users' websites to assess contextual factors. An analysis of the users' personal development records (education, career, interest fields) was conducted in order to validate the qualification profiles of users. Furthermore, we conducted a publication analysis of users' publications to validate the users' contributions to and roles in the RI projects. Market data were used to validate the market success of the RI. Furthermore, internal reports of manufacturers were analyzed in order to validate the interview data. We then discussed the results of the study with the interviewees. Finally, the results of the study were presented and discussed at two international workshops that focused on user innovation research (held at Ludwig-Maximilian University of Munich and at the Sloan School of Management, Boston).

4. Research findings

In presenting the findings, we first provide brief descriptions of the cases. Next, we analyze each case, utilizing the major research questions of our study.

4.1. *Brief case histories*

4.1.1. *Case URS*

In the early 1990s, the German neurosurgeon Volker Urban began questioning conventional technologies for neurosurgery. Based on his experience in the operating room, he realized that a gap existed between the need to operate at sub-millimeter precision and the comparatively low-precision performance of current surgical equipment, and this realization triggered his search for new solutions. In his inventive process, he also looked for technologies outside the medical domain (e.g. nuclear power plants, cockpit technology) that met high-precision requirements. Based

on his exploratory search, Urban developed the vision of utilizing for neurosurgery the new technologies that were incorporated into microcomputers and robotics. He called his vision 'the operating room 2015'. A key element of this vision was a medical robot system for neurosurgery, and a complementary element was an operating cockpit that enabled tactile feedback for the operating neurosurgeon. In September 1995, Urban presented a virtual representation of his concepts at a medical conference entitled, 'Medical goes Electronic'. At the same time, Siemens was preparing for its 150th anniversary celebration and was looking for feasible visions in medical technology that could be presented at this event. Urban contacted Siemens and obtained agreement that this firm would finance the development of a first prototype. However, Siemens had no technological knowledge or core competence to enable it to develop a medical robot at that time. In searching for a suitable technology partner, Urban identified the Fraunhofer Institute in Stuttgart as a worldwide leading competence center in robotics. He contacted the engineers of that institute and convinced them to develop a first prototype. But the entire budget for this project was provided by Siemens. In October 1997, after a year and a half of intensive development and experimentation, the first functioning prototype of a medical robot was presented. Urban then engaged in additional networking activities. First, he identified leading medical institutes at which the medical robot could undergo clinical tests. Second, he actively marketed his concept through Universal Robot Systems (URS), a newly founded firm that was created as a spin-off from the Fraunhofer Institute. Urban became a partner of and scientific advisor for this firm and he promoted the new device at medical congresses and conferences.

4.1.2. Case SPOCS

A major problem in neurosurgery was the inability of surgeons to navigate their instruments in the depths of brain antrums; conventional mechanical procedures could not solve the problem. In the 1980s, the neurosurgeon, Prof. Reinhard, from the University of Basel, who had developed the existing mechanical navigation system, was searching intensively for new solutions to this problem. His solution was to develop the idea of a computer-assisted navigation system that would be capable of guiding the surgeon by providing three-dimensional information in real time. To help realize this solution, Reinhard recruited into

his team a number of neurosurgeons who had the required complementary technological competencies: know-how in computer programming and electronics. Included among his recruits were a neurosurgeon with programming skills and a neurosurgeon with a background in electronics. Reinhard himself was a clock maker before he became a neurosurgeon, and therefore had profound knowledge of fine mechanics. The development team was formed in 1983, and the team completed its first prototype 2 years later without requiring additional external knowledge. In 1985, the first human patient trials were successfully conducted. The new computer-assisted navigation system constituted a radical departure from conventional medical equipment technology in neurosurgery. Precise, controlled navigation in the depths of brain antrums was now feasible, significantly improving the capabilities of neurosurgeons. However, the development team lacked the required market knowledge, financial resources, and marketing and sales know-how to transform the prototype into a marketable product. Consequently, Reinhard contacted Aesculap, a German medical equipment manufacturer that he believed could provide the missing resources. Aesculap had founded a task force in 1987 to explore the potential for using robotics and navigation systems in surgery and was therefore aware of emerging technologies. In 1990, Aesculap finally agreed to cooperate with Reinhard. However, at that time, they were weak in the ability to provide additional technological input for the development of the navigation system because they did not have any specific core competence in the field of sophisticated computer-assisted systems. In 1993–1994, Aesculap put the topic of 'computer assisted navigation system' on hold because of doubts about its market potential, but the topic was back on its agenda by the end of 1994. In 1995, the first navigation systems were introduced to the market.

4.1.3. Case OrthoPilot

In the early 1990s, one of the major problems experienced by orthopedic surgeons was the anatomically accurate positioning of knee endoprostheses. If not positioned accurately, the implants caused severe pain to patients and needed to be replaced after several years. Another major drawback of the conventional procedure was the need for patients to be exposed to radiation-intensive computed tomography scanning in order to obtain images before surgery. These preoperational examinations were time-consum-

ing and expensive, and were not patient compliant. Because of the lack of appropriate equipment, Prof. Saragaglia, an orthopedic surgeon from the University of Grenoble, started to search for new solutions. Based on his knowledge of orthopedics and technological advances in neurosurgery, he developed his vision of a computer-assisted navigation system for orthopedic surgery.

Because he lacked the required technological knowledge, Saragaglia contacted engineers at the University of Grenoble who were experts in the field of programming and identified another orthopaedic surgeon, Frederic Picard, who had a strong background in anatomy. This additional know-how was crucial in developing the required biomechanical solution for the navigation system. As a next milestone, Saragaglia started a project with the goal of developing a first prototype. In 1994, realizing that extensive funding would be required, his first step was to contact the European Union (EU) for financial support. The EU acknowledged the great potential of the idea and agreed to provide kick-off funds. Thus, the project, IGOS (Image Guided Orthopaedic Surgery), was born. It turned out that the initial funds were not sufficient to finance the complete project; consequently, Saragaglia sought out manufacturing firms that would be willing to participate in the project. At that time, the medical equipment manufacturer, Aesculap, was aware of new technological advances in surgery and decided to invest limited funds. In return, Aesculap secured the commercialization rights. With the original funds from the EU and the financial investment of Aesculap, it was finally feasible to develop a first functioning prototype in 1997. After the prototype was successfully tested in 1997, Aesculap assumed management of the entire project. After passing through a number of improvements in its user-friendliness, the navigation system was successfully introduced into the market in 1999. The user-inventors, Saragaglia and Picard, agreed to a licensing contract with Aesculap. Today, the product is a worldwide standard in orthopedic surgery.

4.1.4. Case IMPLANT

Patients of surgeons specializing in hernia repair experienced severe problems after implants were inserted. Patients reported that the implant felt like a 'board'. The medical cause for this effect was that conventional implants were not biocompatible in the sense that enabled optimal tissue regrowth. Conventional implants caused the growth of scar tissue that, in turn, led to the

painful 'board' effect. In 1996, Prof. Schumpelick, a leading surgeon at the University Hospital of Aachen, developed the idea of a completely new implant with biocompatible characteristics. Schumpelick intended to use polypropylene as a new material for the implant and a completely different implant design. The new, lighter material and the new design were expected to support an improved process of tissue regrowth. This idea emerged from Aachen's own research activities at the university labs. As a means of testing the medical relevance of his idea, Schumpelick needed to develop a camera system that verified the specific characteristics of the abdominal wall. Realizing that engineers with specialized know-how in camera technology were needed for the development of the camera system, Schumpelick identified and cooperated with them to develop the new camera system, and together they were finally able to validate his radically new approach successfully. The next milestone was the development of the first prototype, an endeavor that proved to be difficult because Schumpelick lacked the required technical knowledge of textile engineering. Consequently, he contacted the leading institute for textile engineering knowledge, and within a short time the institute developed the first prototypes of the new implant.

In addition, Schumpelick approached a leading manufacturing firm in the field of medical implants, one that had substantial knowledge in the area of biocompatible materials and implants and an excellent reputation in the market. Together, the specialists in camera technology, the research institute in textile engineering, the manufacturing firm, and Schumpelick formed the innovation network for the prospective new implant. Schumpelick not only formed this network but also organized the cooperation within it, including the coordination of the network members' activities, the organization of face-to-face meetings, the creation of status reports, and the setting and realization of milestones. The manufacturing firm successfully tested the biocompatibility of the implant, concluded a licensing contract with Schumpelick, and launched the new implant onto the market in 2002. Today, the implant represents a state-of-the-art technique and is used worldwide for hernia repair surgery.

4.2. Analysis

We now turn to a detailed comparative analysis of these four cases – in particular, analyses of

users as inventors of RI, the entrepreneurial role of innovative users, users as (co)-developers, and benefits for manufacturers.

4.2.1. Users as inventors of RI

In all cases, users were the originators and inventors of RI. The question is what motivated them to do so and what factors enabled them, finally, to develop such innovative concepts. In the beginning, all innovative users shared a major problem: They faced severe difficulties in their day-to-day work that could not be solved by conventional manufacturers' technology or existing medical equipment. For example, the neurosurgeons (cases URS and SPOCS) faced problems of conducting extremely precise (sub-millimeter) work. This need could not be met by standard neurosurgical instruments. The fact that our sample of innovative surgeons encountered the limits of conventional technologies motivated them to search for other, more workable solutions (*motivation induced by problem*). As is known from research on creativity, high problem pressure is a key source for creative activities (Csikszentmihalyi, 1988; Boden, 1994; Collins and Amabile, 1999). In addition to problem-induced motivation, all innovative surgeons were *professionals* in their field and thus had *in-depth knowledge within their domain* of surgery. Furthermore, they had *knowledge about the respective needs* to improve the surgical process. This type of knowledge was gained by extensive learning, experience, and experimentation, and is therefore difficult and costly to transfer to third parties. von Hippel (1994, 1998) calls this knowledge 'sticky'. This type of knowledge served as a crucial basis for the search for solutions that met the specific medical needs.

The idea creation and concept generation processes of the innovative surgeons followed a common pattern in all four cases. The surgeons abstracted from their current use context by searching for appropriate technologies outside of the medical domain. Therefore, *openness to new technologies* was a key prerequisite characteristic that all innovative surgeons shared. Once the relevant technologies were recognized, the innovative surgeons transferred these potential technical solutions to their medical domain. They applied analogical reasoning, which is considered to be a suitable mechanism for the generation of radically new ideas and concepts (Holyoak and Thagard, 1995; Dahl and Moreau, 2002). For example, the neurosurgeon in the URS case looked for solutions to prevent trembling of the

neurosurgeon's hands and thus to realize precision in the sub-millimeter area. In his search for solutions, the neurosurgeon observed nuclear power plants, for example. Employees in nuclear power plants need a transmitter between them and the fuel elements. This observation triggered the idea that a neurosurgeon could also use a transmitter between his hands and the patient to control trembling more effectively. Following this analogy, he closely investigated various principles of kinematic solutions. Robotic systems are based on kinematics; consequently, the concept of a medical robot arm for neurosurgery was born.

In addition to problem-induced motivation, openness, and prior knowledge, a number of enabling factors played a major role in the development of the radically new concepts. Taking a closer look at these enabling factors, two types of innovative users could be distinguished. The first type was embedded into a context with close *access to interdisciplinary know-how*. These users were surgeons at university hospitals that were affiliated with technical universities or that had access to departments of technical universities. This interdisciplinary context inspired the surgeons' creative thinking as state-of-the-art technologies could be experienced. According to the concept of 'absorptive capacity' (Cohen and Levinthal, 1990), access to interdisciplinary know-how served to increase the creative capacity of these users. Another important contextual factor of this user type was the availability of *resources for research* (time, funds, personnel). These resources enabled the innovative surgeons to perceive technologies outside of the medical domain and to think about possible technology transfers.

The second type of innovative user did not have access to these supportive factors. However, this type exhibited a strong *intrinsic motivation*. In addition to high problem pressure, these surgeons regarded the search for radically new solutions as their hobby and spent a substantial amount of spare time on it. This finding is in line with research on creativity that highlights the importance of intrinsic motivation for highly creative activities (Csikszentmihalyi, 1988).

Would manufacturing firms have been capable of developing the radical new concepts of innovative users? This question, by nature, is difficult to answer. It should be noted, however, that the development of these concepts required a deep understanding of user needs and creative incorporation of relevant technologies outside of the medical domain. The in-depth knowledge about user needs is difficult to transfer to manufacturing

firms (von Hippel, 1994, 1998). Consequently, users have almost exclusive access to this type of knowledge. This reasoning might explain why users rather than manufacturing firms developed these radically new concepts.

4.2.2. Entrepreneurial role of users

Besides their inventor role, the surgeons performed an entrepreneurial role as innovation network builders in the RI projects. They established and managed the networks required to transform their radically new concepts into first prototypes and later into marketable products. The innovative surgeons concerned themselves with the need for sufficient financial resources, and invested significant time and effort in convincing potential sources to provide such funds. They recruited the talents of experts from research institutes to provide the necessary technological know-how and expertise. Furthermore, they interacted with manufacturers to convince them to realize the implementation of their concepts. Figure 1 illustrates this entrepreneurial role.

A closer look into the structure of the user-initiated innovation networks reveals that these networks had a low degree of density or interconnectedness and that the user-inventors were

highly central within the networks. The observation of the user-inventors' networking activities raises the question: why did they assume such a challenging and - role? Our analysis reveals that the users' entrepreneurial role can be explained as a combination of several factors. First, these users faced severe problems in their work that could not be solved through conventional technology. The high problem pressure was the driving force not merely for the development of concepts but also for the formation and management of the innovation networks. Second, the users developed the concepts without any involvement of third parties such as technological experts or manufacturers. Hence, the users were the original inventors and thus had conceptual solutions for their problems already at hand. Third, in this early phase of the RI projects, potential manufacturers were not willing to adopt the users' radically new concepts, perhaps because, in all cases, the projects did not fit with the core competencies and technology strategies of the manufacturing firms. Looking back at the radical nature of these concepts in their early stage, this is not surprising. In those early days, an adoption of the users' concepts by a manufacturer would have required it to build the necessary technological competencies, mostly from scratch. Furthermore, the manufacturers

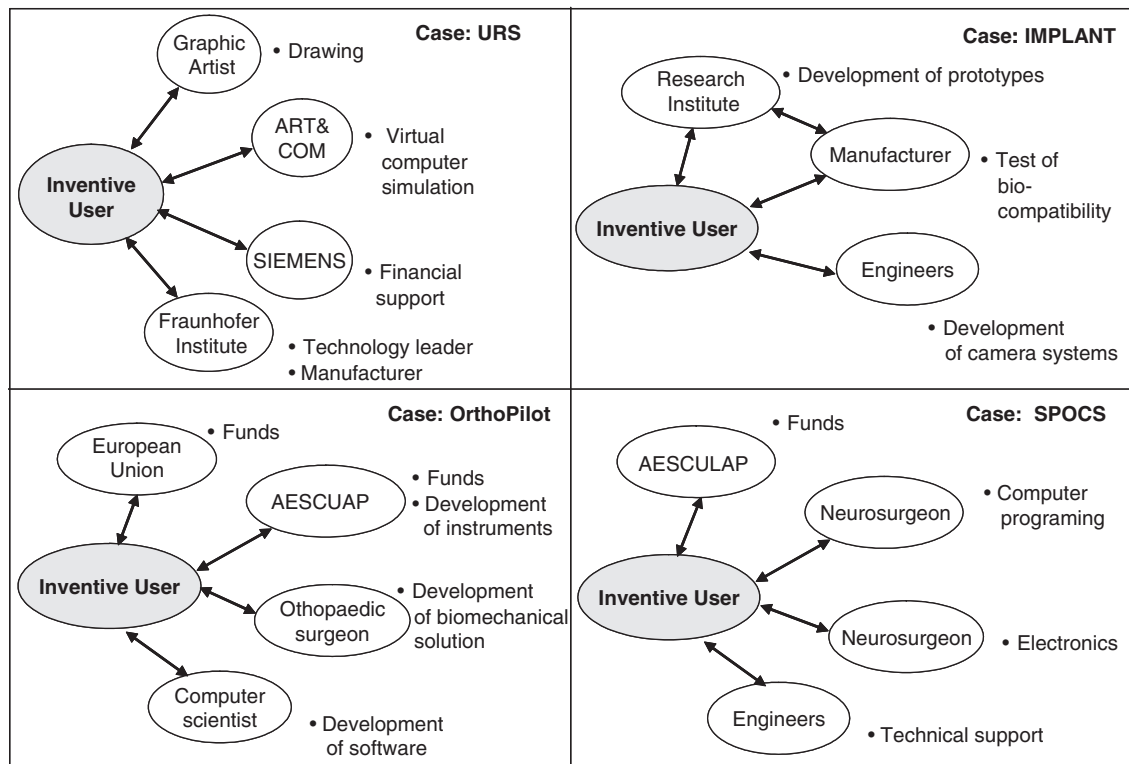


Figure 1. Entrepreneurial role of user-inventors.

were deterred by the high technological and market uncertainties associated with these innovations. They were confronted with a situation in which the medical application of the respective technologies was just emerging and that the volume, market share, and growth rate of prospective markets were as yet unknown. Therefore, the degree of innovation is a key factor that explains why manufacturers did not become actively involved in the early phase of the innovation projects and why the user-inventors had to fill this gap. Because the manufacturers did not adopt the innovative concepts, the user-inventors needed to take over an entrepreneurial role, at least temporarily. Fourth, the user-inventors, of course, did not have all the needed competencies and lacked the resources required to transform their concepts into prototypes and marketable products. As for missing competencies, users lacked the technological competencies and marketing knowledge that are required in order to implement RI into the market successfully (Leonard-Barton and Doyle, 1996; O'Connor, 1998; McDermott, 1999; O'Connor and Veryzer, 2001). This required set of competencies exceeds the 'bounded rationality' of a single user or even a team of users (Simon, 1957; March, 1978; Simon, 1982). In addition to a lack of some competencies, the innovative users did not have all the necessary financial, human, and marketing resources required to develop prototypes and marketable products. For this reason, the innovative users were dependent upon external support from tech-

nological experts, development partners, and medical equipment manufacturers. Research on nascent entrepreneurs highlights a lack of competencies and resources as important antecedents for networking (e.g. Birley, 1985; Davidsson and Honig, 2003). In summary, a high problem pressure, a conceptual solution at hand, a reluctant position of manufacturers (mainly caused by the high degree of innovativeness of the product or project), as well as missing competencies and resources were antecedents of the observed networking activities of innovative users (Figure 2).

4.2.3. Users as (co)-developers

In addition to networking, the user-inventors took over another typical function of manufacturers. They became *developers* or *co-developers*. What characteristics enabled them to perform this role? A case comparison of user characteristics and associated development contributions implies a swell model with separate layers. Each layer can be considered as a critical activity level whereby higher layers are associated with more ambitious and challenging contributions. The first layer consists of passive development contributions in the user domain. These contributions include the articulation of needs and/or problems and the evaluation of concepts and prototypes. Being the standard routine of ordinary users in incremental innovation projects, our analysis reveals that in RI projects even this type of contribution requires distinct user characteristics. The cases show that users need a *motivation caused by a current*

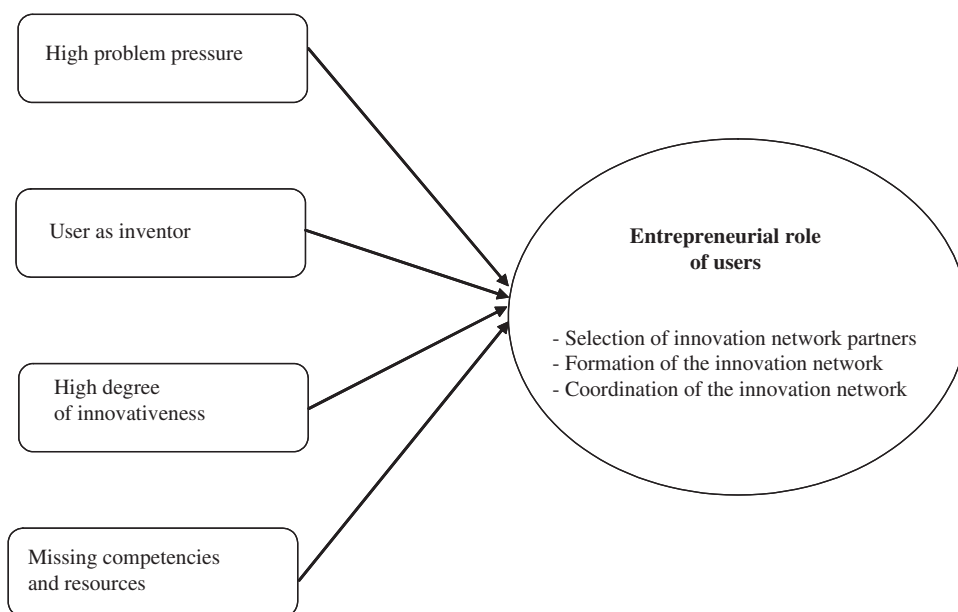


Figure 2. Antecedents of the entrepreneurial role of users.

problem, by *openness to new technologies*, and by *imagination capabilities*. This finding can explain why the so-called opinion leaders are not necessarily suitable informants in RI projects. Opinion leaders may lack 1 of these 3 prerequisites. Particularly, openness to new technologies is not necessarily associated with opinion leaders as their status is often based on conventional technologies. In the URS and IMPLANT cases, opinion leaders were indeed opponents of the RI, and were therefore incapable of validly evaluating the concepts and prototypes. Our observation confirms results from former empirical studies showing that opinion leaders are not necessarily capable of recognizing the benefit of RI in the prototype stage (Lynn et al., 1996; Salomo et al., 2003).

The next layer constitutes active development contributions in the user domain, contributions that involve the development of user's own solutions. The case studies reveal that users need an additional set of characteristics in order to perform well on this layer. First, users need to have a *high competence in their own domain*. In order to develop solutions for RI, one needs to have a profound understanding of the elements, the causes, and the effects of a certain domain. Second, users must have a *tolerance for ambiguity* or the ability to handle uncertainty. In the present situation, the characteristic means that users who actively engage in the development of RI must be able to handle a great deal of uncertainty regarding the final output and benefit of their development efforts. Third, users need to have *access to technological know-how*. This resource is critical for providing immediate feedback to innovative users regarding the technological feasibility of their solutions. Using an iterative process, innovative users can leverage the feedback to improve their own solutions (Thomke, 1998, 2003; Thomke and von Hippel, 2002). In the OrthoPilot case, an innovative surgeon developed the biomechanical solution of the computer-assisted navigation system for orthopedics. By having access to the technological knowledge computer science possessed by the co-developing engineer, the innovative surgeon was able to improve his biomechanical solution iteratively. Fourth, users need *resources for their own research activities*. One explanation of why this contextual factor is critical at this layer is the high complexity of such a task. Users must deal intensively with the specific subject at hand and therefore require intellectual free space and resources such as time, facilities, and funds. In summary, specific characteristics

enable users to realize active development contributions in their own domain.

To reach the highest layer, active development contributions in the technological domain, users also need *technological competencies*. The SPOCS case illustrates this point. In this case, a team of innovative neurosurgeons developed not just the concept but also a first prototype of a computer-assisted navigation system for neurosurgery. This development was possible as, in combination, the surgeons possessed all the necessary complementary technological knowledge on mechanics, computer programming, and electronics. One surgeon was a professional watch maker before he embarked on his educational track for neurosurgery. He therefore had the technological know-how associated with mechanics. Another surgeon trained himself auto-didactically in computer programming until he had sufficient computer programming skills. A third team member had a strong background in electronics. Obviously, users with a diverse set of technological capabilities ('cross-qualification') form an attractive group as development partners for RI. Our analysis reveals that users who are capable of performing on this highest layer of the swell model are the exception rather than the normal case. Users primarily focused their development activities on the user domain. One explanation for this observation is rooted in the nature of RI, particularly in a high-technology field. The innovations profiled in our case studies incorporate new and highly complex technologies and a separate educational track is required in order to develop sufficient technological know-how to master them. Most surgeons do not have the time and the 'absorptive capacity' to build up such technological competencies. The swell model with its three distinct layers is illustrated in Figure 3.

The swell model provides a framework that explains the characteristics that enable distinct development contributions by users in RI projects. Our analyses regarding the surgeons' active development contributions reveal that surgeons applied knowledge that they already had at hand rather than actively searching for new knowledge. This finding confirms evidence from user innovations in sporting equipment (Luethje et al., 2005) and in open source software (Raymond, 1999).

4.2.4. Benefits for manufacturing firms

Manufacturing firms that adopted the users' concepts and prototypes in later phases of technological product development benefited from the

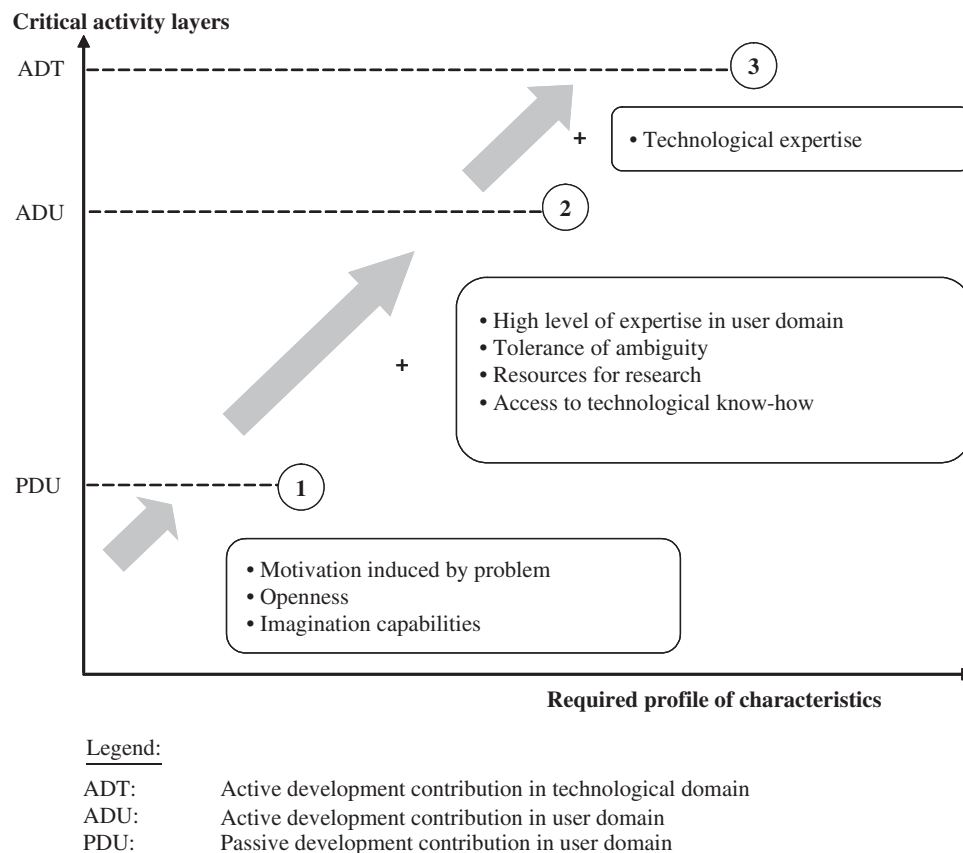


Figure 3. Swell model of development contributions by users for radical innovations.

users' contributions on 3 dimensions: the product innovation, the process, and the technology.

4.2.4.1. Product innovation dimension. By linking up with innovative users, manufacturers learned about concepts that implied a significant change of the 'standard of care' in the addressed medical field. At that stage, leading manufacturers of medical equipment were not capable of developing these concepts themselves, as the concepts departed dramatically from their core competencies. Only by making close contact with the small circle of highly creative surgeons were medical equipment manufacturers in a position to *absorb these radically new concepts*. By cooperating with these surgeons, manufacturers also learned to *improve the quality* of the RI *substantially* (in terms of performance improvements). This benefit was derived from the surgeons' active development contributions in the medical and technological domains alike.

4.2.4.2. Process dimension. Manufacturers benefited from innovative surgeons with respect to the efficiency of the innovation process. First, the

development contributions of the surgeons enabled manufacturers to *decrease development time and development costs*. In all four cases, the innovative surgeons dominated large parts of the innovation processes, enabling manufacturers to take over complete prototypes. Therefore, manufacturers did not have to invest in the cost-intensive early phases of the RI process and were able to introduce the medical devices into the market in comparatively short times. Second, the innovation-related information provided by the surgeons *enhanced the decision quality* of the manufacturers' decision process. Based on the surgeons' information, the manufacturers were able to select the most promising prototype versions.

4.2.4.3. Technology dimension. Manufacturing firms that linked up with the innovative surgeons also realized benefits on the technology dimension. First, manufacturers were able to *acquire new technological competencies*. The transfer of technological knowledge from surgeons to manufacturers occurred in two different modes. In a direct mode, innovative surgeons had technologi-

cal competencies and were therefore in a position to transfer this knowledge to the manufacturers' engineers (as in the SPOCS case). In an indirect mode, users identified leading research institutes and developed their first prototypes with these institutes. By taking over these prototypes, manufacturers were able to acquire the underlying technology (cases OrthoPilot, URS, IMPLANT). Second, manufacturing firms learned about *relevant technological trends* in the medical equipment industry by cooperating with the innovative surgeons. As the surgeons' concepts and prototypes contained leading-edge technology for the specific medical field, manufacturers were alerted to technological trends and received 'early signals' of emerging technological discontinuities. The technological trends were robotic systems in surgery, computer-assisted navigation systems, and bio-compatible implants. Third, by linking up with the user-initiated networks, manufacturers learned about *leading technological research institutes* relevant for RI in their business divisions and were able to *extend their innovation and technology-related networks*.

5. Discussion and conclusion

Our findings have implications for user and open innovation research as well as for corporate practice. Our case study analysis reveals that the profile of users who are in a position to contribute substantially to the early phases of RI projects differs significantly from those users who are typically involved in conventional marketing research. Is the lead user concept a suitable approach? With regard to this question, it is important to note that the innovative users in our case samples do not exactly meet the classical lead user definition. The needs that the user innovators faced were commonly faced by the mass of surgeons in the respective surgical fields. The user innovators therefore did not exhibit trend leadership with respect to market needs. However, the innovative users in our case sample share some characteristics that are associated with lead users. First, the innovative users were highly motivated to develop new solutions. Second, in 2 cases, the innovative users were neurosurgeons who can be categorized as extreme users as they faced the need for extremely high precision. It has been shown that lead users frequently appear among 'extreme users' (von Hippel et al., 2000; Lilien et al., 2002). We therefore conclude that the innovative users in our sample have close simi-

larities with lead users. However, our analysis reveals that additional user characteristics are needed in order to contribute substantially to the development of RI. Such characteristics include being open to new technologies, being embedded in a supportive environment, and having strong intrinsic motivation.

Even though the innovative users do not exactly meet the classical lead user criteria, they can be considered as lead users in a different sense. These innovative users recognized the relevance and benefit of new technologies far earlier than did manufacturing firms and peer users. They identified the technological opportunities because of their specialized knowledge and experience (Shane, 2000). These new technologies became the medical standard in many different medical applications. Consequently, the innovative users can be regarded as 'technology lead users' – trend leaders on the technological dimension. The identification of these 'technology lead users' is a crucial task for firms that seek to benefit from users in the early stages of RI projects. Furthermore, 'technology lead users' can be leveraged for purposes of technology forecasting (Gerybadze, 1994) in manufacturing firms. Our study therefore emphasizes the relevance of users from a technological perspective. In general, the user innovation approach – and the lead user approach, in particular – have been mainly discussed as tools to identify market needs and market trends (Luethje and Herstatt, 2004). Our study sheds light on the technological dimension of user innovations and highlights some users as a valuable resource for the identification of technological trends.

The observation, in all four cases, that users dominated large parts of the innovation process, shows that the 'customer active paradigm' (von Hippel, 1976, 1979) can also occur in RI projects. In particular, the observed entrepreneurial role of user-inventors is an interesting aspect for further (user) innovation research. Studies in different settings are needed to learn more about its antecedents and implications. Furthermore, the user-initiated networks should be studied in more detail, utilizing the 'network theory of stakeholder influences' and social network analysis (Rowley, 1997) in which dimensions such as network density or centrality of the user innovator must be analyzed with a goal of deepening our understanding of user-driven innovation networks.

Finding that a high degree of innovativeness is an important antecedent of the observed entre-

preneurial role of users can also be analyzed from a dominant design theory perspective (Abernathy and Utterback, 1975, 1978; Utterback and Suarez, 1993). On the macro level, RI constitutes technological discontinuities in an industry and renders existing dominant designs obsolete. They therefore initialize a phase of 'ferment', in which high uncertainties exist with respect to the emerging dominant design (Anderson and Tushman, 1990, 1991). Owing to these design instabilities, established manufacturing firms are reluctant to invest in new product development in this phase of the industry life cycle. In contrast, users might derive a direct benefit from the new technologies by way of tailored solutions for their needs. They may thus derive innovative and, because of the reluctance of manufacturers, entrepreneurial benefits. Design instabilities caused by technological discontinuities therefore might be an antecedent for user-driven innovations. Further research is needed in order to understand more clearly the technological discontinuities and design instabilities as contingency factors for the sources of innovation in general and user innovations in particular.

Regarding research on open innovation, our study highlights users as a co-source of RI outside established manufacturing firms and raises questions about the requisite 'openness' (Laursen and Salter, 2005) and competencies established firms need in order to benefit from the creative and managerial potential of entrepreneurial users. Users who act in an entrepreneurial fashion allow firms to outsource, at least partially, the R&D and marketing function. If some users are indeed able to dominate the innovation process of RI, we might need to rethink the conventional wisdom that considers that manufacturers dominate the RI process and involve users as passive, residual entities. In our case sample, we found that users involved manufacturers in the development process at a time in which they lacked the required competencies and resources to proceed on their own. This observation suggests a need for further research in order to develop characteristics of capable and 'open' manufacturers as suitable cooperatives or partners for entrepreneurial users. Manufacturers need to establish an ability to differentiate between promising and less promising user innovations. In this connection, a conceptualization and empirical test of a 'user involvement competence' of manufacturing firms and its impact on corporate success is an interesting avenue for future research. Transferring the lead user concept to the manufacturer side leads

us to think about 'lead manufacturers' that recognize the potential of emerging technologies earlier than others and therefore have an absorptive capacity for the contributions of entrepreneurial users.

Our study also raises questions about the adequacy of the concept of core competencies (Prahalad and Hamel, 1990), in cases for which users and other actors outside of established manufacturing firms are the source of RI (Christensen, 2005). In the observed cases, established firms nearly rejected successful RI because of their narrow focus on core competencies. Future empirical research on open innovation must therefore investigate the adequacy and change in the concept of core competencies in the context of open innovation. In an open innovation scenario, the networking competence of firms (Ritter, 1999; Ritter and Gemuenden, 2003) might become an increasingly important core competence.

With respect to corporate practice, we recommend that manufacturing firms systematically leverage capable users for RI projects. The involvement of such users turned out to be an effective learning mechanism to enhance a firm's RI capability in this study. One important dimension of the organizational competence for RI, therefore, is the capability of identifying this group of highly creative and entrepreneurial users. In this context, the identified characteristics of innovative users can be applied by manufacturers as a search grid in order to identify highly creative users more systematically. The identification of such users can increase the creative capacity of a manufacturer as radically new ideas and concepts can be gained. Similarly, the swell model can be used as a search grid for the identification of such users, ones that can play the role of (co)-developers for RI projects. The swell model implies a need for conceptual thinking in 'user pyramids'. We assume that the number of users decreases with higher layers as higher layers are more challenging and require additional characteristics. Consequently, we contend that it is far more difficult for manufacturing firms to identify users on top of the pyramid (users capable of active development contribution in the technological domain) than to identify users at the bottom of the pyramid (users capable of passive development contributions in the user domain). As few users are capable of delivering substantial contributions for RI, manufacturing firms need to conduct the user selection process diligently. In this connection, a pyramiding approach⁵ of market research was shown to be suitable for the

identification of leading-edge expertise within a given population (von Hippel et al., 2005). Once identified, manufacturers need to cooperate intensively with those highly creative users in order to learn from them in an effective way. The identification of users who take over large parts of the development process is a particularly suitable strategy for small- and medium-sized firms to use in developing RI under R&D-budget restrictions.

In the cases involved in our study, manufacturing firms did not become active in the development process until the first prototypes had been developed. Although firms were aware of the radically new approaches and technologies, they did not adopt them in early stages. Was the 'wait & watch' behavior of the manufacturing firms the proper reaction? In order to answer this question, the success of the radical innovations needs to be assessed. In three cases, the associated new products turned out to be successful both on the technological and market dimensions. Consequently, our results reveal that manufacturers were initially reluctant to enter concepts that subsequently turned into successful innovations. However, this does not imply that the initial reluctance of manufacturers was not the proper reaction. Instead, the manufacturers minimized the risk of their own investments by this passive behavior. As the benefits of the RI became more apparent, the manufacturers took over complete prototypes from the innovative users and were able to introduce the RI to the market in a relatively short time. In the URS case, the innovative user became a partner of a research institute's spin-off entity. The innovation turned out to be a market failure. Established manufacturers did not engage in this RI project; consequently, their assessment mechanisms appear to have worked well in this case.

The observed networking activities of user-inventors also have implications for corporate practice. For manufacturing firms, it becomes crucial to identify those user-initiated innovation networks. By linking the firm into these networks, manufacturers can participate in RI projects and ensure access to the exclusive rights for commercialization. This is a useful approach to reduce investments and risks in RI projects. Furthermore, manufacturing firms can identify leading research institutes for their RI work and extend their innovation networks. The recommendation for manufacturers is to create an organizational unit that systematically scans the relevant environment for those users who work on the leading

edge of their disciplines and for innovation networks that emerge from these users' activities.

To recapitulate, our study sheds light on the user-manufacturer interface in those RI projects that are originated by users. It should be noted, however, that our study was exploratory in nature and contained only four RI projects. Further research is needed to strengthen our grounded theory. A critical question is: can the identified patterns be observed in other industries? The technology in the medical equipment industry has specific characteristics that limit the generalization of our results. One important speciality of this industry is that users (in our cases, surgeons) are professionals; they use medical equipment technology as part of their profession. Professional users can apply their professional knowledge (in our cases, in-depth medical knowledge and experience in a specific discipline of surgery) to develop innovative solutions for their needs. This characteristic might place them at an advantage in making contributions to RI projects, as opposed to users who handle products or technologies as part of their hobby (e.g. in sports like mountain biking and kayaking). Further research is needed to extend our study to other industries and to test our exploratory models by using large-scale, quantitative studies.

Acknowledgements

We gratefully acknowledge the helpful comments and thoughts provided by Eric von Hippel, Carliss Baldwin, and Christian Luethje.

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Notes

1. Users can be independent individuals or firms (von Hippel, 2005).
2. In this connection, it should be noted that manufacturing firms also develop innovations primarily with a low-to-medium degree of innovativeness.
3. For a review of these studies and a discussion on how to improve the efficiency and effectiveness of the lead user method, see Luethje and Herstatt (2004).
4. In this connection, it needs to be taken into account that the RI projects had great significance for the innovative users and the manufacturing firms. Therefore, they were well recalled by both parties.
5. 'Pyramiding' means that the market researcher asks an informant within a defined population whether he or she knows someone who is highly knowledgeable about a specific field of expertise. Once identified, this person is asked whether he or she knows someone who is even more knowledgeable. This procedure is repeated until the top experts are identified (von Hippel et al., 2005).

Appendix 1

Table A1. Degree of innovativeness

(1) Market dimension
New benefit for the user
Higher benefit for the user
Degree of newness to users/Familiarity of users with technology
Degree of required behavior change for users
Degree of required learning by users
Degree of change in market structures
Degree of creation of new markets
Potential for competitive advantage
Degree of market uncertainty for manufacturers
Uncertainty about market acceptance
Uncertainty about market potential
Uncertainty about market growth
(2) Technological dimension
Degree of newness of technological principle
Degree of complexity of technology
Degree of technological uncertainty for manufacturer
Uncertainty about feasibility of the technology
Uncertainty about the development time
Uncertainty about the development costs
(3) Organizational dimension
Degree of newness to the manufacturer
Degree of required change in production process
Degree of required change of competencies
Degree of required change in organizational culture
Degree of required change in organizational structure
Degree of required change of (human) resources
Degree of required change in strategy
(4) Environmental dimension
Creation of a new infrastructure
Alterations in regulation required in order to implement innovation
Critical debate of innovation in society

Appendix 2

Table A2

Table A2. Selected coding tables

User characteristics	Indicators	Examples from interview protocols
Motivation induced by problem	Perceived problem	'We had the problem that patients would feel very uncomfortable with the implant after a while'. (Informant: User/Case: IMPLANT)
	Dissatisfaction with existing solutions	'The surgeons in pediatrics were very dissatisfied, because there were no adequate radiology systems for them'. (Informant: R&D/Case: GCF)
Intrinsic motivation	Fun	'To me it was fun to work on my vision of the operating room 2015. This is nothing you do because someone tells you to do it. You do it because you enjoy it'. (Informant: User/Case: URS)
	Hobby	'The whole thing was a hobby for him. He spent a tremendous amount of his spare time to bring his vision and ideas to life'. (Informant: Marketing/Case URS)
Openness to new technologies	User deals with new technologies (e.g. by reading technical journals)	'I am constantly dealing with new technologies that might be relevant for my medical field. I study technical journals to keep me posted about what is going on with respect to new technologies'. (Informant: User/Case: IMPLANT)
	Openness to accept new technologies as solutions for the medical field.	'Neurosurgeons are from their attitude much more open to new technologies like orthopedics... the reason is that they need to work with a precision in the sub-millimeter area and conventional technologies did not really meet this need'. (Informant: R&D/Case: SPOCS)
Technological competencies	Background in science or technical disciplines	'Prof. Reinhard was remarkably capable in the technical field. He had profound know-how in the field of mechanics, as he took an education in watch making before starting a career in medicine'. (Informant: R&D/Case: SPOCS)
	Software-programming skills	'Dr. Horstmann has written the software programs for the device. He acquired these skills autodidactically'. (Informant: R&D/Case: SPOCS)
Resources for research	Team	'This surgeon has a team of surgeons who do research'. (Informant: Marketing/Case: IMPLANT)
	Research laboratories	'Prof. Reinhard had his own laboratories and technical equipment like milling machines'. (Informant: R&D/Case: SPOCS)
	Budget	'We had our own funds and raised funds for our innovation project'. (Informant: User/Case: OrthoPilot)
	Time	'The innovative activities of the users in Basel were supported from Swiss public funds'. (Informant: Marketing/Case: SPOCS)
Access to interdisciplinary/ technological know-how	Short communication distance to interdisciplinary expertise	'These users at university hospitals also have the time to work on innovative solutions. This is a clear difference from surgeons in ordinary hospitals'. (Informant: Marketing/Case: IMPLANT)
	University hospital is embedded in a university of technology	'They also work interdisciplinarily in this university hospital. There are several technical research institutes close by; this way there were different disciplines involved and they had a short communication distance to communicate and exchange ideas'. (Informant: R&D/Case: IMPLANT)
	Departments of the university of technology are embedded in the university hospital	'This group of users has the advantage of being embedded in the university of technology'. (Informant: R&D/Case: IMPLANT)
		'... and so the Lab for Robotics and Navigation was born inside the hospital; it was located in the department of the hospital'. (Informant: User/Case: OrthoPilot)

Table A2. (Contd.)

User contributions and roles	Indicators	Examples from interview protocols
Independent development of concept [inventor role]	User develops concept without external assistance	'Dr. Urban has developed this concept. He had the idea right from his work in the operating room in his mind and then elaborated on it until the concept was there'. (Informant: Marketing/Case: URS)
Passive development contribution in user domain [(co)-developer role]	Definition of requirements Information how to integrate the innovation into the use context	'The user in Munich has defined the profile of requirements for his specific medical application field'. (Informant: R&D/Case: SPOCS) 'The observations in the operating room were always very valuable. The surgeons showed us how to integrate the robot into the surgical process'. (Informant: R&D/Case: URS))
Active development contribution in user domain (development of solutions in the medical domain) [(co)-developer role]	Development of the medical solution for the innovation	'Dr. Pickard developed the solution for the anatomical and biomechanical part. These were the medical foundations of the OrthoPilot'. (Informant: R&D/Case: OrthoPilot))
Active development contribution in technological domain (development of solutions in the technological domain) [(co)-developer role]	Development of components of the prototype/development of the complete prototype Computer programming	'The surgeons in Basel developed the complete first prototype by themselves'. (Informant: R&D/Case: SPOCS) 'This surgeon has developed the software for the navigation system'. (Informant: R&D/Case: SPOCS)